



DIGITAL VLSI DESIGN

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PART 1

MOS Inverters (Static Characteristics)

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MOS Inverters (Static Characteristics)

Part d - Inverters with n type MOSFET load

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Inverter with n-type MOSFET Load

A inverter circuit with active load instead of load resistive is used in nmos inverter. The active load used in nmos Inverter may be

- I) Enhancement nmos load - These are referred as enhancement load nmos inverter
- II) Depletion nmos load – These are referred as depletion load nmos inverter.

Advantages:

- a) Silicon area occupied by transistor is usually smaller than resistive load.
- b) The overall performance is better compared to resistive load.

Why V_{out} is limited to $VDD - V_{T,load}$?

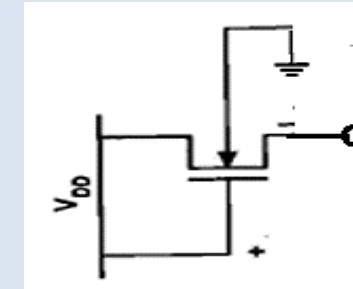
E-nmos as a pass transistor can pass strong logic 0 but not strong logic 1

When source voltage become

$V_S = VDD - V_{T,load}$, then

$V_{GS} = V_{GG} - V_S = VDD - VDD + V_{T,load}$

$V_{GS} = V_{T,load}$.



$$V_{GS} = VDD - V_S$$

As a results e-nmos load gets into cut off and then source of load is at $VDD - V_{T,load}$

How to get $V_{out} = V_S = VDD$?

A separate source is used for Drain and gate.

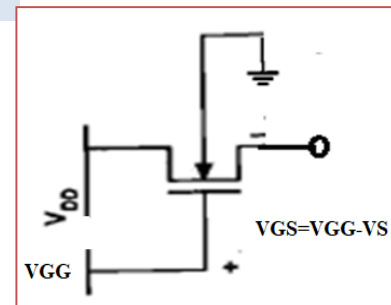
In this V_{GG} is set such that device is in condcuts and the output at the source is equal to VDD .

$V_{GG} = VDD + V_{T,load}$

$V_S = VDD$, then

$V_{GS} = V_{GG} - V_S = (VDD + V_{T,load}) - VDD$

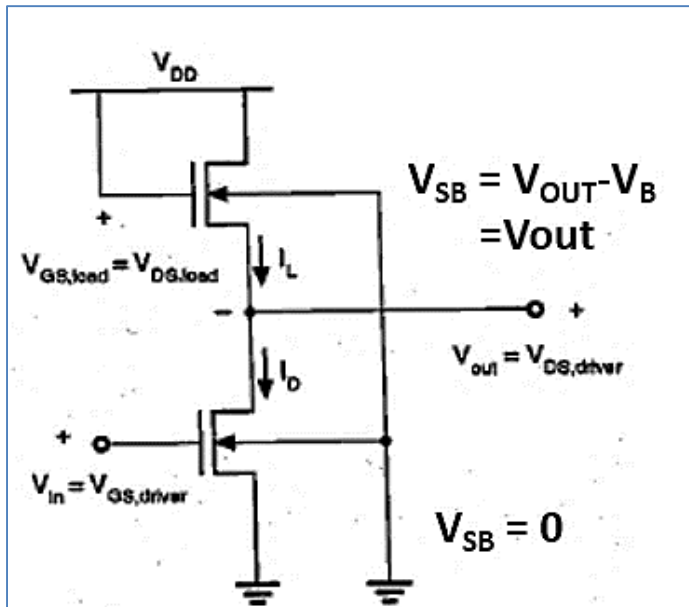
$V_{GS} = V_{T,load}$.



Enhancement load nmos Inverter

In this enhancement nmos load is used. Based on bias voltage applied to gate of e-nmos load, the load nmos can be operated either in saturation region or in linear region. As a result we have two possible for enhancement load nmos inverter

- a) Saturated enhancement type nmos load inverter
- b) Linear enhancement type nmos load nmos inverter



$$V_{T,load} = V_{T0,load} + \gamma \left(\sqrt{|2\phi_F| + V_{out}} - \sqrt{|2\phi_F|} \right)$$

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b) Linear enhancement type nmos load nmos inverter

In this there is a separate supply for drain and gate of load e-nmos.

But V_{GG} which is set such that the device always operates in Linear region and gives $V_{out} = V_{OH} = V_{DD}$ when V_{in} to driver transistor is $V_{in} < V_{T0, driver}$

$V_{in} = 0$, $V_{out} = V_{OH} = V_{DD}$

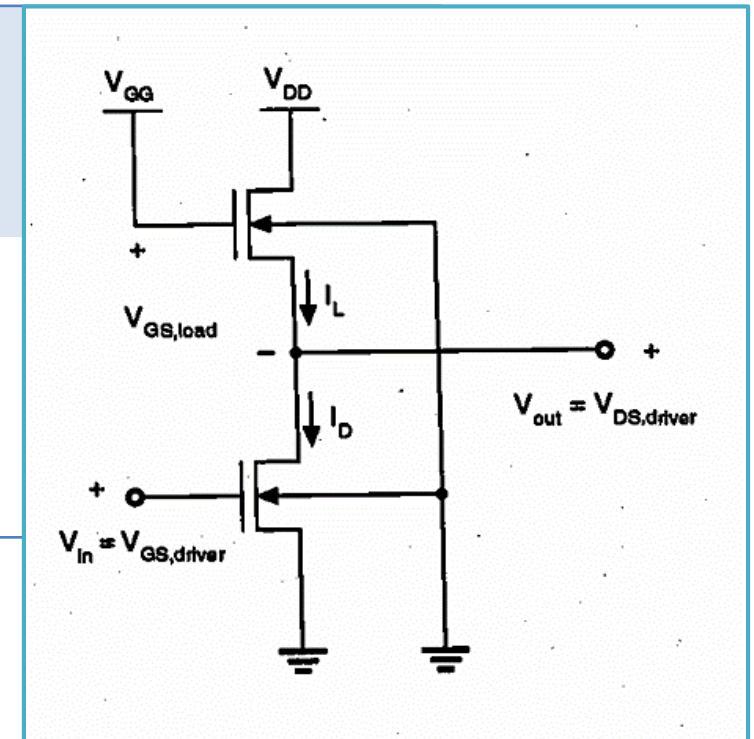
$V_{GS, load} = V_{GG} - V_S = V_{DD} + V_{T, load} - V_{DD} = V_{T, load}$

$V_{GS} \geq V_{DS} + V_{T, load} = V_{T, load}$

$V_{in} = V_{DD}$, $V_{out} = V_{OL} = 0$

$V_{GS, load} = V_{GG} - V_S = V_{DD} + V_{T, load} - 0 = V_{DD} + V_{T, load}$

$V_{GS} \geq V_{DS} + V_{T, load} = V_{DD} + V_{T, load}$



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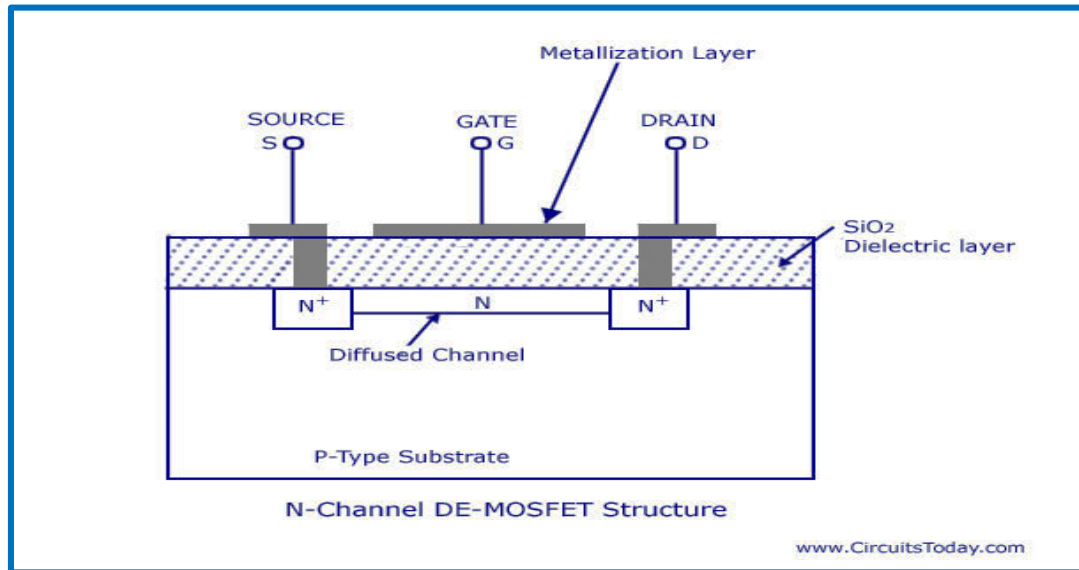
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NMOS Depletion Load Inverter

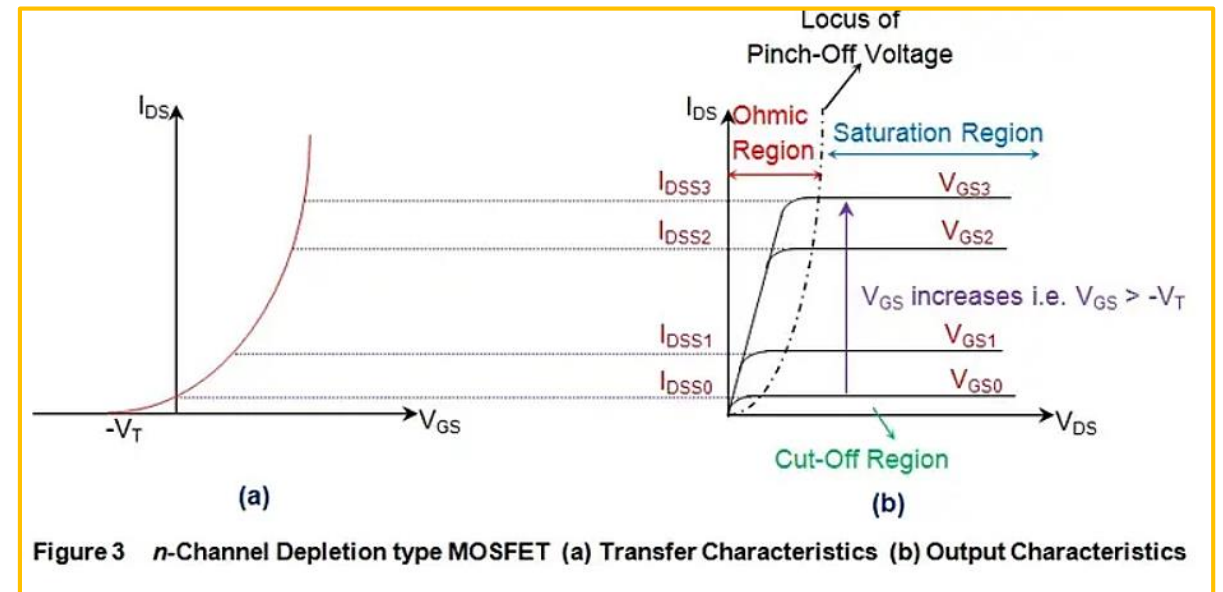
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Depletion type nmos device

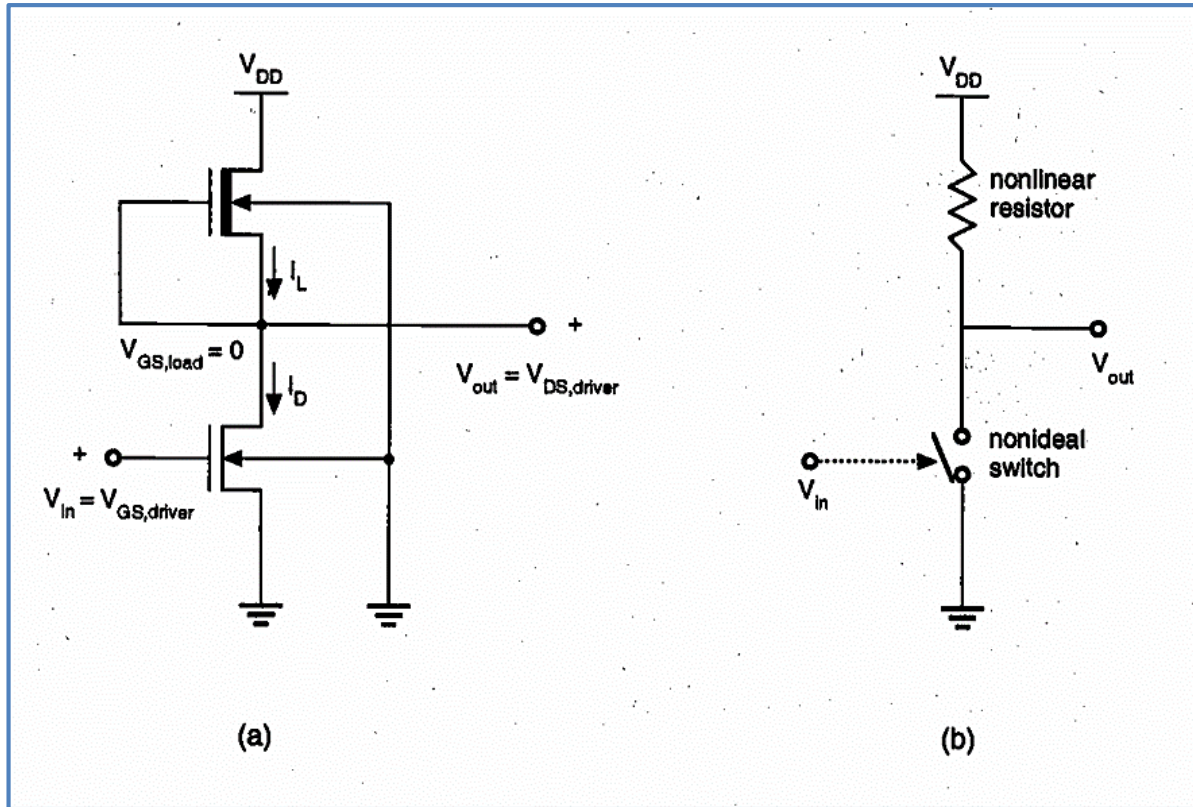


If $V_{GS}=0$, Device is conducting with $I_D=I_{D0}$



MOS Inverters (Static Characteristics)

Depletion type nmos Inverter

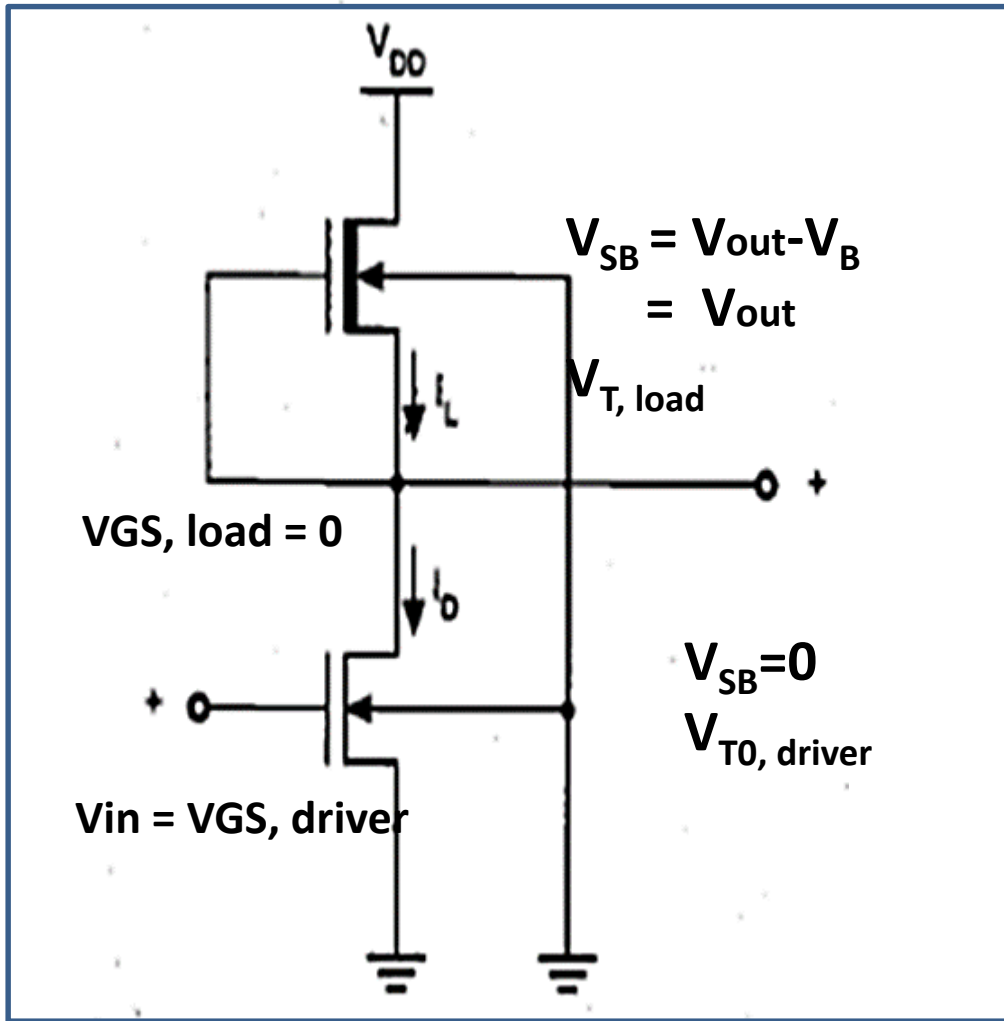


(a) Inverter circuit with depletion-type nMOS load. (b) Simplified equivalent circuit consisting of a nonlinear load resistor and a nonideal switch controlled by the input.

The immediate advantages of implementing this circuit configuration are:

- (i) sharp VTC transition and better noise margins,
- (ii) single power supply, and
- (iii) smaller overall layout area.

Depletion type nmos Inverter



$$V_{T,load} = V_{T0,load} + \gamma \left(\sqrt{|2\phi_F| + V_{out}} - \sqrt{|2\phi_F|} \right)$$

Device in Saturation:

$$I_D = \frac{kn}{2} (V_{gs} - V_{tn})^2$$

$$I_{D,load} = \frac{k_{n,load}}{2} \cdot [-V_{T,load}(V_{out})]^2 = \frac{k_{n,load}}{2} \cdot |V_{T,load}(V_{out})|^2$$

Device in Linear:

$$I_D = \frac{kn}{2} [2(V_{gs} - v_{tn})V_{ds} - V_{ds}^2]$$

$$I_{D,load} = \frac{k_{n,load}}{2} \cdot [2|V_{T,load}(V_{out})| \cdot (V_{DD} - V_{out}) - (V_{DD} - V_{out})^2]$$

MOS Inverters (Static Characteristics)

Depletion type nmos Inverter

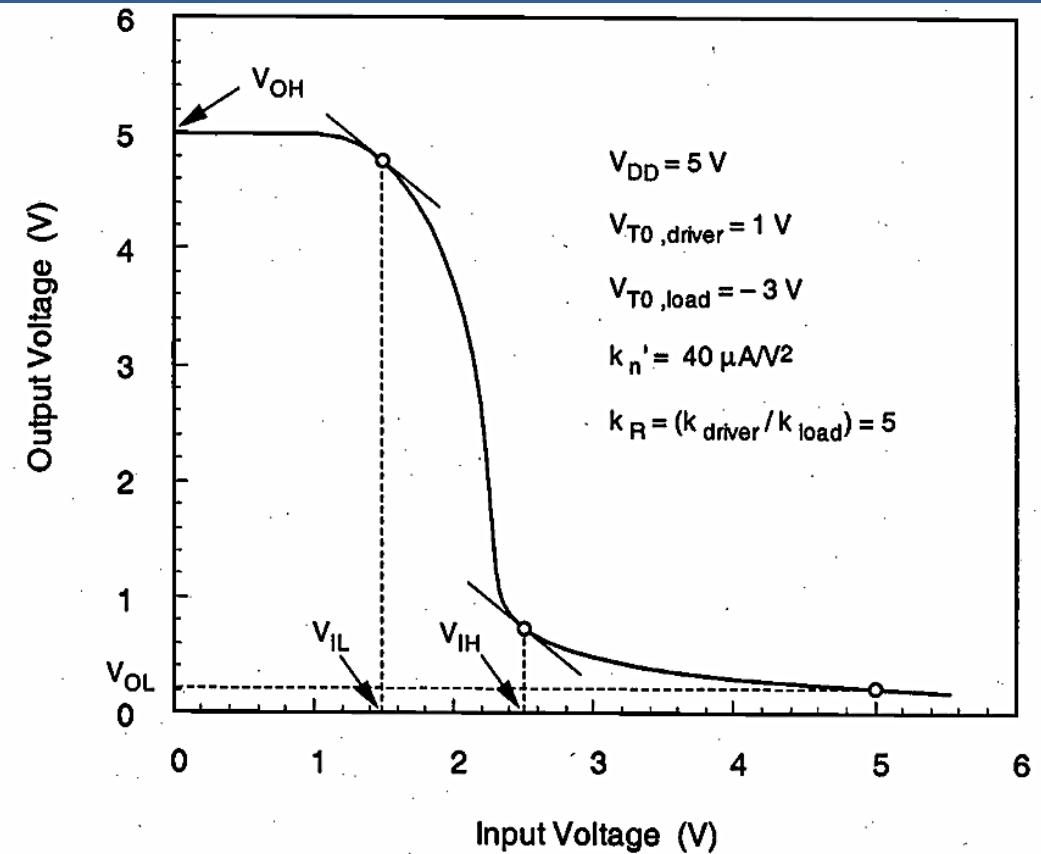
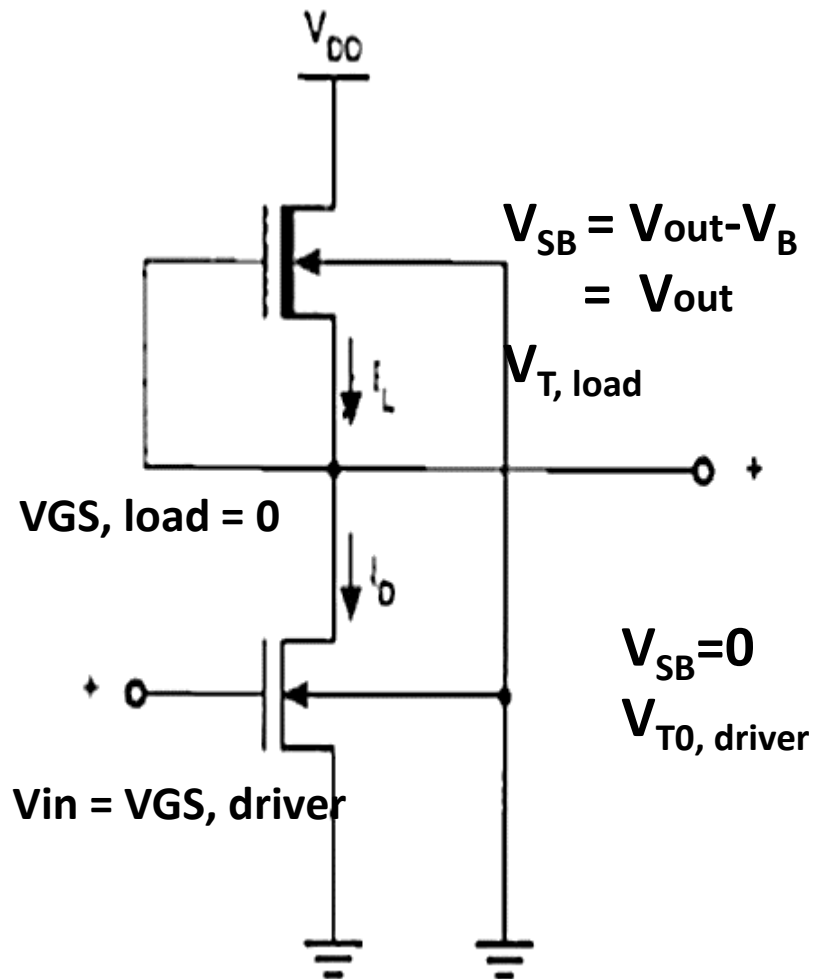
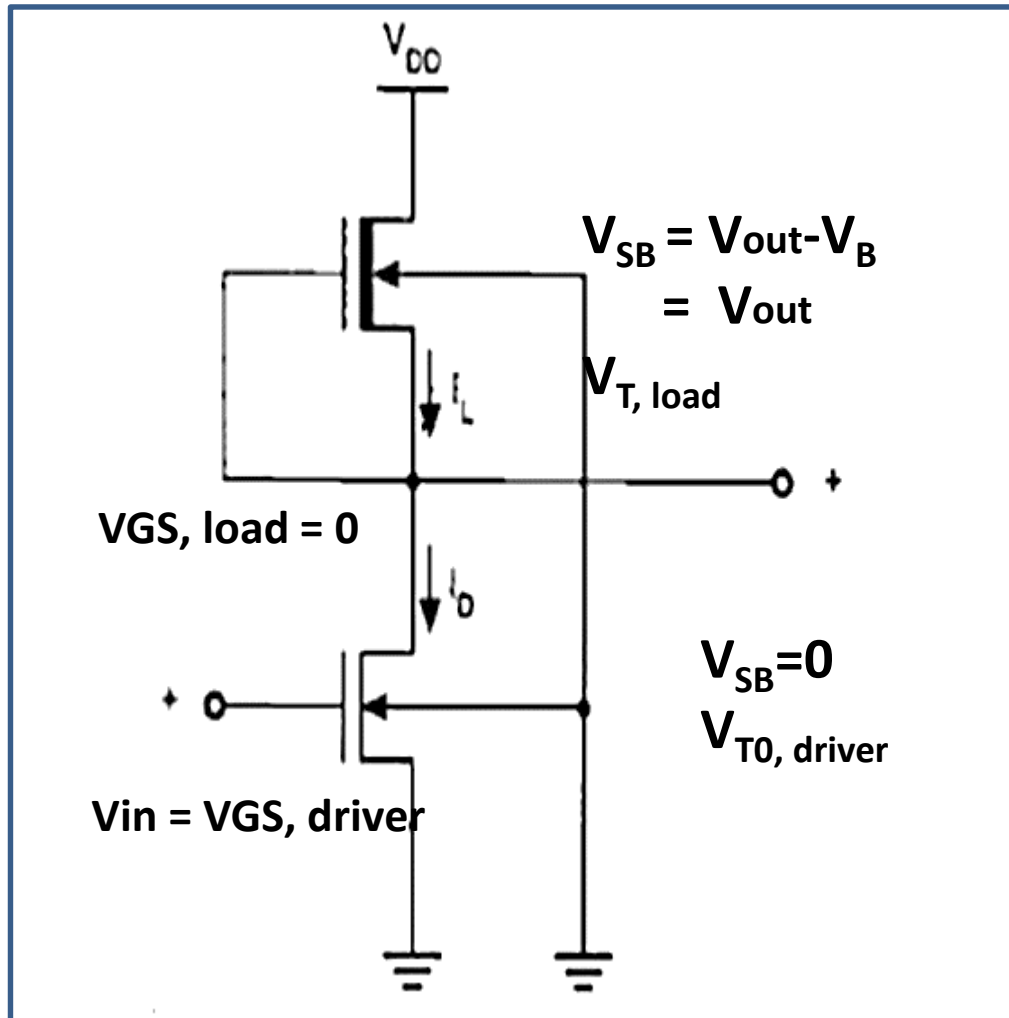


Figure 5.13. Typical VTC of a depletion-load inverter circuit.

MOS Inverters (Static Characteristics)

Depletion type nmos Inverter



Note:

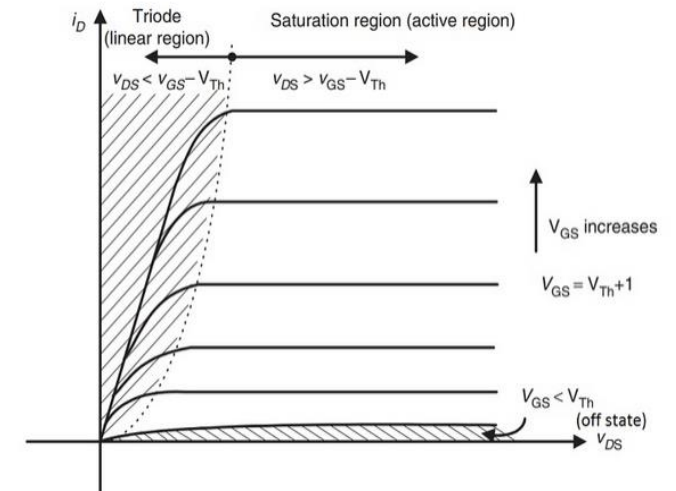
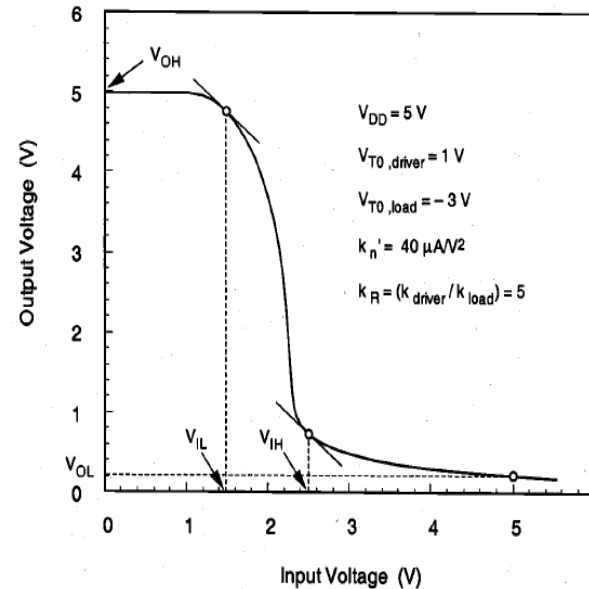
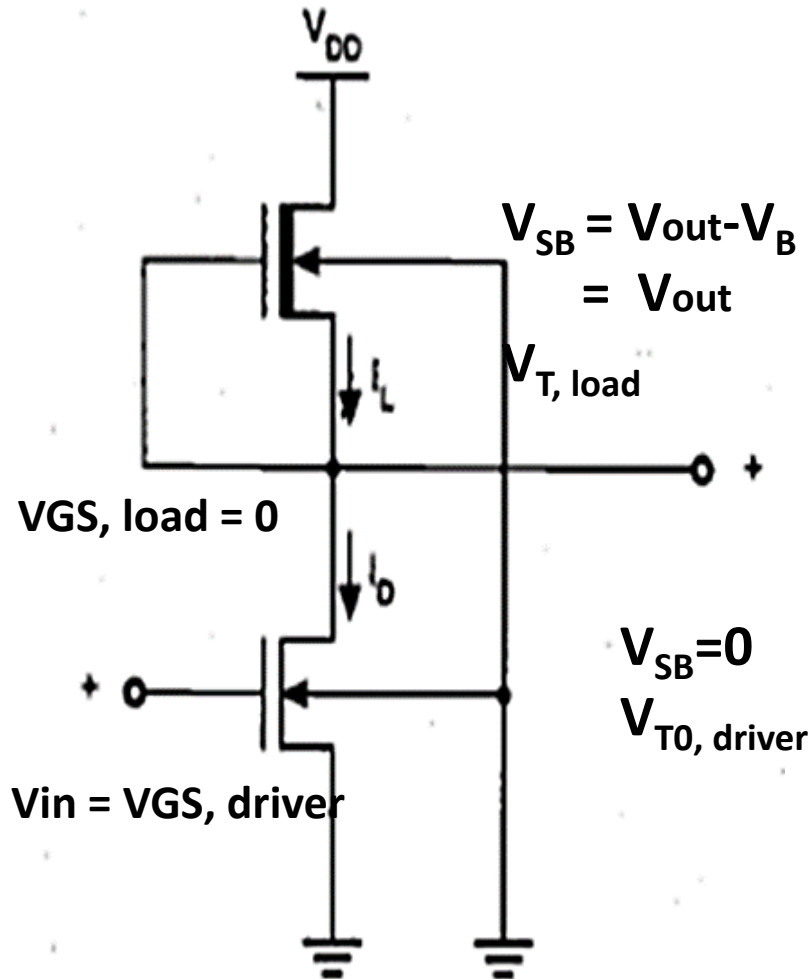
The fabrication process for producing an inverter with an enhancement-type nMOS driver and a depletion-type nMOS load is slightly more complicated and requires additional processing steps, especially for the channel implant to adjust the threshold voltage of the load device.

The resulting improvement of circuit performance and integration possibilities, however, easily justify the additional processing effort required for the fabrication of depletion load inverters.

MOS Inverters (Static Characteristics)

Depletion type nmos Inverter

Note: Design of Depletion load nmos Inverter is not covered from the syllabus



V_{in}	V_{out}	Driver operating region	Load operating region
V_{OL}	V_{OH}	cut-off	linear
V_{IL}	$\approx V_{OH}$	saturation	linear
V_{IH}	small	linear	saturation
V_{OH}	V_{OL}	linear	saturation

MOS Inverters (Static Characteristics)

Power Calculation

Power : $P_{DC} = V_{DD} \cdot I_{DC}$

If the duty cycle of input pulse is 50% then

$$P_{DC} = \frac{V_{DD}}{2} [I_{DC}(V_{in} = Low) + I_{DC}(V_{in} = High)]$$

In depletion load inverter, When $V_{in} = Low$, Driver nmos is OFF ($I_D = 0$), $V_{out} = V_{OH}$. There is no significant current flow through the load and the driver transistor. When $V_{in} = High$ (V_{DD}), both driver and load transistor are ON.

$$P_{DC} = \frac{V_{DD}}{2} [0 + I_{DC}(V_{in} = High)]$$

$$\begin{aligned} I_{DC}(V_{in} = V_{DD}) &= \frac{k_{load}}{2} \cdot [-V_{T,load}(V_{OL})]^2 \\ &= \frac{k_{driver}}{2} \cdot [2 \cdot (V_{OH} - V_{T0}) \cdot V_{OL} - V_{OL}^2] \end{aligned}$$

$$P_{DC} = \frac{V_{DD}}{2} \cdot \frac{k_{load}}{2} \cdot [-V_{T,load}(V_{OL})]^2$$



THANK YOU

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